Industrial Silicon Fundamental Analysis

A Focus on Chinese Market



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Chapter 1

Supply Landscape Overview

1.1 What Is Industrial Silicon?

Industrial silicon, also known as metallurgical-grade silicon (MG-Si), is a refined form of elemental silicon with a typical purity of 98.5%∼99.9%. It is produced by carbothermic reduction of high-purity quartz (SiO_2) in submerged arc furnaces at temperatures exceeding 1,700°C. The key chemical reaction is:

$$SiO_2 + 2C \rightarrow Si + 2CO \uparrow$$

Here, carbon is provided by reductants such as petroleum coke, coal, or wood chips.

Table 1.1: Classification of Industrial Silicon by Grade

Grade	**Typical Purity (% Si)**	**Impurity Thresholds (Fe/Al/Ca)**	**Typical Application**
553#	98.5∼98.7	Fe $\leq$ 0.5%, Al $\leq$ 0.5%, Ca $\leq$ 0.3%	Aluminum alloys, steel deoxidizers
441#	98.9∼99.1	Fe $\leq$ 0.4%, Al $\leq$ 0.4%, Ca $\leq$ 0.1%	Automotive casting, rubber, magnesium alloys
421#	99.2∼99.4	Fe $\leq$ 0.4%, Al $\leq$ 0.2%, Ca $\leq$ 0.1%	Polysilicon feedstock, organosilicon production
3303#	$\geq$ 99.3	Fe $\leq$ 0.3%, Al $\leq$ 0.3%, Ca $\leq$ 0.03%	Chemical-grade silicon (PV/electronic use)
2202#	$\geq$ 99.4	Fe $\leq$ 0.2%, Al $\leq$ 0.2%, Ca $\leq$ 0.02%	High-purity solar-grade feedstock
97#	97.0–98.5	Non-standardized	Low-end metallurgy, basic alloying

Key downstream applications of industrial silicon include:

- **Polysilicon production (solar-grade silicon):** The largest and fastest-growing demand segment, used in photovoltaic (PV) cells.

- **Organosilicon compounds:** Used in chemicals, lubricants, sealants, and cosmetics.
- **Aluminum-silicon alloys:** Enhances strength and thermal performance in automotive and aerospace castings.

Why a Futures Market? Industrial silicon is a high-energy, high-capex commodity with significant price volatility due to cyclical demand (e.g., solar installations), energy cost shocks, and policy shifts (e.g., anti-overcapacity campaigns). The introduction of silicon futures allows:

- Producers and end-users to hedge against price risk.
- Investors to express views on macro/policy-driven supply-demand imbalances.
- Improved price discovery and forward guidance for the industry.

China is the world's dominant producer and consumer of industrial silicon, with over 80% of global capacity located in the country as of 2024. The Shanghai Futures Exchange (SHFE) launched silicon futures in 2022, quickly gaining traction due to strong industry participation.

1.2 Global Capacity Overview

Global industrial silicon capacity reached **8.81 million metric tons** in 2024, up **3.2%** year-over-year, according to the China Photovoltaic Industry Association (CPIA). **China remains the dominant producer**, accounting for **82.6%** of global capacity in 2024. This reflects a further gain in global share, driven by **272,000 tons** of new domestic capacity, largely from deferred 2023 projects and policy-supported vertical integration. Capacity additions outside China remained minimal.

Major global producers include China, Brazil, the United States, Norway, France, Russia, and Germany. However, China's dominance continues to reshape the global supply landscape, particularly as downstream demand becomes increasingly solar-driven.

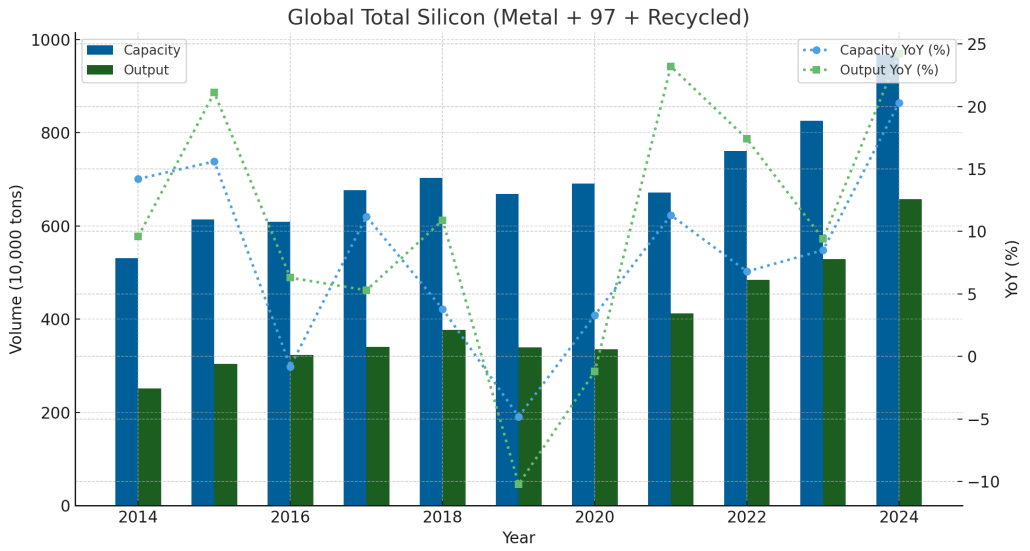


Figure 1.1: Global Silicon Capacity–Output Trend

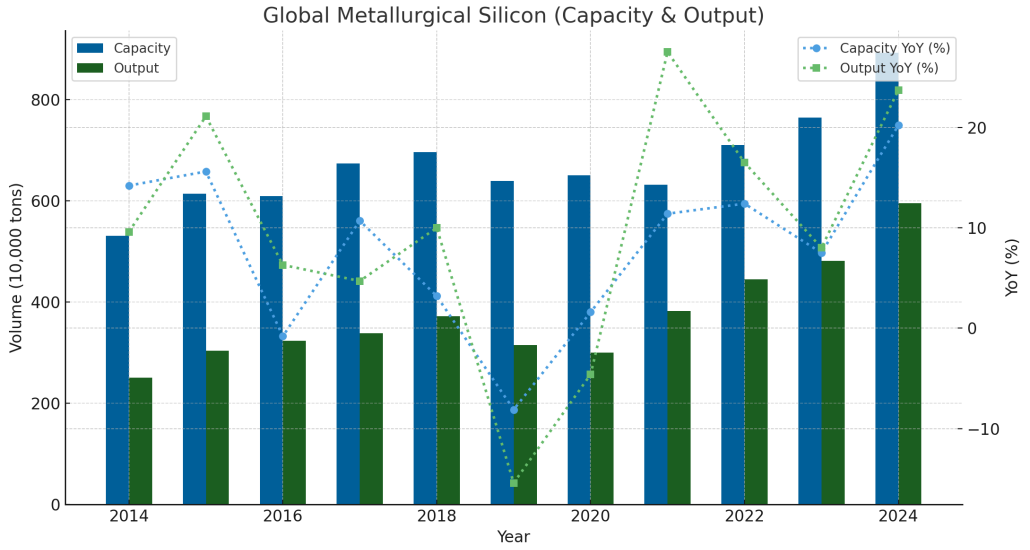


Figure 1.2: Global Industrial Silicon Capacity–Output Trend

1.3 China: Regional and Corporate Distribution

1.3.1 Total National Capacity and Output

As of the end of 2024, China's total industrial silicon capacity stood at **7.14 million tons**, continuing a three-year growth trend. Output reached **4.9 million tons** in 2024.

The growth is largely attributed to:

- Successful commissioning of new projects from previous years.
- Productivity upgrades in existing facilities through technological renovation.
- Increased participation in the futures market, boosting demand for deliverable-grade products.

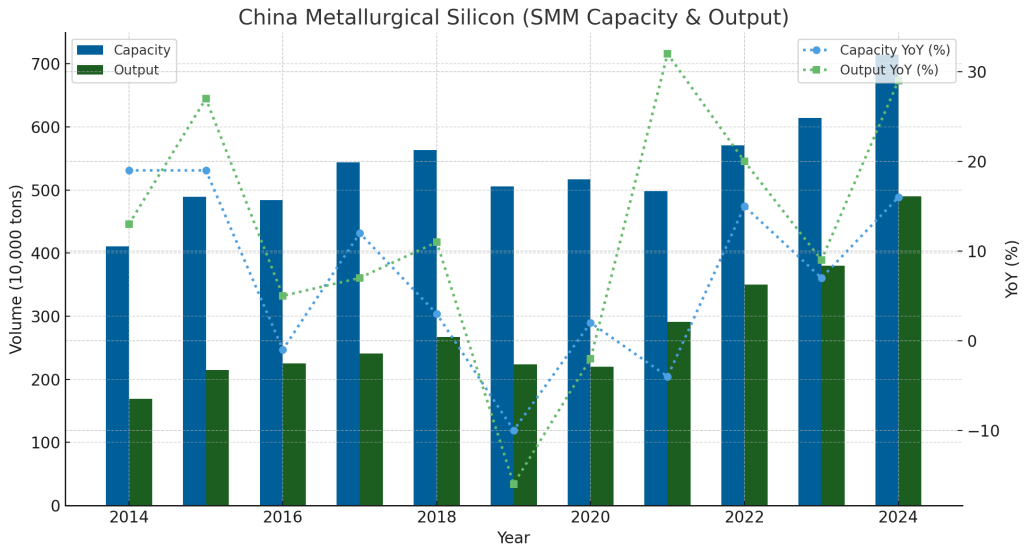


Figure 1.3: China Industrial Silicon Capacity-Output Trend (2014–2024)

1.3.2 By Region: Xinjiang, Yunnan, Sichuan

Industrial silicon is a highly electricity-intensive industry, with power accounting for **30∼40%** of total production costs. As such, electricity pricing and power structure are critical determinants of regional competitiveness. China's capacity is concentrated in two core clusters: **Xinjiang** in the Northwest, and **Yunnan and Sichuan** in the Southwest. These regions differ in energy source, seasonality, product quality, and policy orientation.

Xinjiang

Home to approximately **38∼40%** of national capacity, Xinjiang benefits from low-cost, stable coal-fired power (0.26∼0.33 Yuan/kWh), making it the dominant base-load supplier. However, widespread use of all-coal reductants leads to higher impurity levels (Fe, Al, Ca), producing mostly **553# metallurgical-grade silicon** used in aluminum and steel.

Yunnan

Yunnan contributes **22∼25%** of capacity, driven by seasonal hydropower. Utilization peaks during the wet season (May–October), when power prices fall below 0.2 Yuan/kWh. Use of wood chips and petroleum coke as reductants results in **higher-purity outputs**, suitable for **421# and 3303#** chemical-grade silicon and polysilicon feedstock.

Sichuan

With **12∼14%** of capacity, Sichuan shares similar seasonality and product orientation with Yunnan, focusing on chemical-grade production. Several producers here are integrated into downstream organosilicon chains, improving value capture.

Policy support for *green silicon + downstream upgrading* has reinforced the strategic role of the Southwest, especially amid China's decarbonization push.

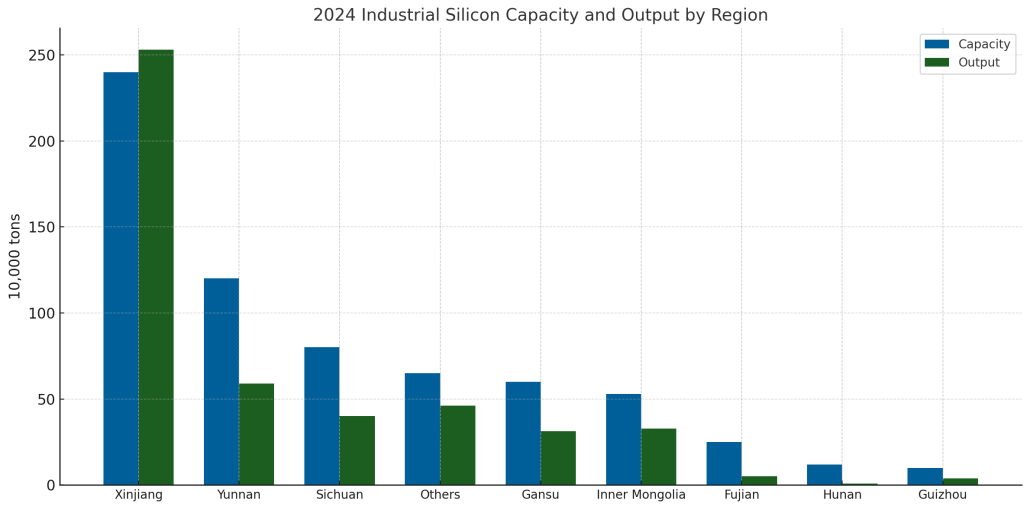


Figure 1.4: China Industrial Silicon Regional Distribution (2024)

1.3.3 By Producer: Major Industrial Silicon Companies

The Chinese industrial silicon industry currently exhibits a structurally fragmented supply side in terms of **nominal capacity**, but a relatively concentrated landscape in terms of **actual output**. As of mid-2024, the top five producers (CR5) collectively account for nearly **35% of total national designed capacity**, yet contribute an estimated **60∼65% of actual output**. This discrepancy reflects the sharp divergence in utilization rates across producers: large players operate integrated, energy-secured, and policy-supported facilities, while smaller producers face greater exposure to market fluctuations and regulatory tightening.

Despite an influx of announced or ongoing furnace projects in Inner Mongolia, Gansu, and Ningxia, a significant share of new capacity faces commissioning delays or suboptimal utilization. According to public project disclosures and company reports, over 800–1,000 thousand tons of capacity was under construction or announced in 2024. However, due to persistently low industrial silicon prices (hovering around the cost line), **many new entrants find it financially unsustainable to commence production**. Even established producers report profit compression, with spot prices approaching or falling below the full-cost breakeven level (estimated at 10,000∼12,000 Yuan/ton for medium-cost producers). As a result, capacity expansion is likely to decouple from actual supply growth in the short term, reinforcing the production dominance of CR5 firms.

Table 1.2: Major Industrial Silicon Producers in China (2025)

Company	**Main Regions**	**Monthly Output (thousand tons)**
Hoshine	Xinjiang, Yunnan	181
East Hope	Gansu, Qinghai	66
GCL	Inner Mongolia, Xinjian	36
Tongwei	Sichuan	28
TBEA	Xinjiang, Inner Mongolia	14

1.4 Industry Structure Evolution and Capacity Outlook for 2025

The most notable structural shift in China's industrial silicon sector is the emergence of **vertically integrated production chains**, a defining feature of the industry's current development phase. Leading producers such as **Hoshine** and **Tongwei** not only operate large-scale silicon smelting capacity but are also tightly linked to downstream segments: Hoshine in organosilicon, Tongwei in polysilicon and wafers. This model ensures upstream supply stability and downstream cost control, and is increasingly favored in project approvals and capital allocation. Integration reduces market risk, enhances pricing power, and aligns closely with the government's strategic goal of building **high-quality, stable, and controllable industrial supply chains**.

The shift toward "high-quality" is also shaped by China's tightening environmental regulations. Industrial silicon production is energy-intensive and generates toxic exhaust gases, making environmental compliance costly. As policymakers strengthen enforcement—particularly around *exhaust gas treatment*, *furnace efficiency*, and *carbon intensity*, those smaller, decentralized producers face escalating barriers to entry. These environmental demands have raised the R&D and operational thresholds, giving large, integrated firms with stronger technical capacity a distinct advantage.

Geographically, capacity expansion has concentrated in **Northwest China**, notably Xinjiang, Gansu, Qinghai, and Inner Mongolia, where favorable electricity tariffs, abundant industrial land, flexible environmental enforcement, and strong labor supply support large-scale development. In contrast, power-constrained provinces like Yunnan and Sichuan have seen a relative decline in new project approvals. As clustering deepens, a new round of **infrastructure and policy co-location** (e.g. captive power, rail access, tax rebates) is reinforcing a capital-intensive, cost-optimized silicon belt across the Northwest.

Another visible trend is the widespread adoption of **33,000 kVA or larger submerged arc furnaces** in new builds, reflecting a clear push toward scale efficiency. These units have significantly lower energy consumption per ton of silicon and are better suited for integration with modern environmental control systems. Combined with the broader trends of regional clustering and integration, this signals a fundamental transition: China's industrial silicon industry is moving decisively toward **large-scale, capital-intensive, and policy-aligned production ecosystems**, concentrated in the Northwest.

Chapter 2

Cost Structure in Major Production Regions

2.1 Cost Structure Analysis

The production of industrial silicon is fundamentally a reduction reaction involving quartz (SiO_2) and a carbonaceous reductant, typically coke or coal. The simplified stoichiometric equation is:

$$SiO_2 + 2\,C \rightarrow Si + 2\,CO$$

Based on molar masses (SiO_2: 60.1 g/mol, C: 12.0 g/mol, Si: 28.1 g/mol), the theoretical mass input required to produce 1 ton of silicon is approximately **2.14 tons of silica** and **0.86 tons of carbon**. However, actual industrial production involves a series of inefficiencies due to heat loss, side reactions, impurity removal, and incomplete reduction, leading to significantly higher real-world consumption.

In practice, the **actual unit material consumption** typically reaches **2.6∼2.8 t/t of silica**, **1.6∼2.0 t/t of reductant**, and **12,000∼12,500 kWh/t of electricity**. In addition, each ton of silicon produced requires approximately **0.08∼0.10 t/t of graphite electrode (ordinary power)**. These input coefficients form the basis of a cost model built on a "*unit price* × *unit consumption*" framework. This modeling approach allows for a clear decomposition of regional and temporal cost differences, as variations in electricity tariffs, raw material prices, and furnace efficiency directly impact each cost component. Therefore, an accurate estimate of marginal cost begins with a bottom-up assessment of material and energy inputs per ton of silicon.

The total production cost per ton of industrial silicon (C_{total}) can be modeled as:

$$C_{\text{total}} = C_{SiO_2} + C_{\text{C}} + C_{\text{electricity}} + C_{\text{electrode}} + C_{\text{labor}} + C_{\text{other}}$$

where each component is calculated as:

$$C_i = q_i \times p_i$$

Here, q_i denotes the actual unit consumption (e.g., tons or kWh per ton of silicon), and p_i is the unit price of each input. C_{other} accounts for taxes, depreciation, amorti-

zation, maintenance, auxiliary materials, and so on. If we are talking about C_{cash}, then depreciation and amortization are not included.

Among all input factors, electricity remains the most cost-sensitive component in industrial silicon production. While the industry-wide average electricity consumption typically ranges between **12,000∼12,500 kWh/t**, large-scale integrated producers with captive power plants and waste heat recovery systems can reduce consumption to as low as **10,000 kWh/t or even below**. These firms benefit not only from improved furnace efficiency but also from access to internal electricity generation at significantly lower tariffs, further amplifying cost savings.

For all key inputs—electricity, silica, reductants, electrodes—cost control depends on two critical levers: **unit consumption** and **unit price**. Leading firms are able to optimize both simultaneously: they adopt advanced process controls to minimize material usage, and leverage economies of scale or local policy advantages to secure preferential input pricing. For instance, a 0.05 Yuan/kWh difference in electricity tariff can result in over 600 Yuan/t variation in total production cost, assuming average consumption of 12,000 kWh/t. **As a result, actual production costs can vary substantially across firms and regions, making accurate cost assessment particularly challenging.**

In the following section, we analyze the cost structures of representative firms (**Hoshine**) and benchmark regions (**Xinjiang** and **Yunnan**), to provide a grounded understanding of cost variability in the current market.

2.2 Regional Cost Approximation

2.2.1 Xinjiang

Table 2.1: Cost Structure of Industrial Silicon at Xinjiang (Average)

Cost Item	Unit Consumption	Unit Price (Yuan)	Cost (Yuan)	Share (%)
Direct Raw Materials			**3,815**	**41.27**
Silica (delivered)	2.7	400	1,080	11.68
Coke (washed)	1.65	900	1,485	16.06
Wood chips	0.6	500	300	3.24
Graphite electrode	0.1	9,500	950	10.28
Electricity	11,500	0.32	**3,680**	**39.81**
Others			**1,000**	**10.82**
Transportation			500	5.41
Packaging/sealing			50	0.54
Equipment maintenance			350	3.79
Warehousing			30	0.32
Inbound inspection			70	0.76
Labor			**400**	**4.33**
Environmental Fees			**200**	**2.16**
Desulfurization			120	1.30
Dust removal			30	0.32
Solid waste disposal			50	0.54
Sales/Admin/R&D			**150**	**1.62**
Total Cost (Incl. Tax)			**9,245**	

Hoshine (Shanshan Base 2024)

Table 2.2: Cost Structure of Industrial Silicon at Hoshine (Shanshan Base)

Cost Component	Unit Consumption	Unit Price (Yuan)	Cost (Yuan)	Share
Direct Raw Materials			**3,620**	**43.61%**
Silica	2.7 t/t	400	1,080	13.01%
Coke (washed)	1.65 t/t	900	1,485	17.89%
Wood Chips	0.4 t/t	500	200	2.41%
Graphite Electrode	0.09 t/t	9,500	855	10.30%
Electricity	11,000 kWh/t	0.28	**3,080**	**37.11%**
Labor			**400**	4.82%
Other Expenses			**1,000**	12.05%
Environmental Fees			**200**	2.41%
Total Cost (incl. tax)			**8,300**	

2.2.2 Sichuan / Yunnan

Wet Season

Table 2.3: Cost Structure of Industrial Silicon in Southwest Wet Season

Cost Component	Unit Consumption	Unit Price (Yuan)	Cost (Yuan)	Share (%)
Direct Raw Materials			**4,860**	**46.46**
Silica	2.5 t	550	1,375	13.15
Coke	0.7 t	1,800	1,260	12.05
Petroleum Coke	0.8 t	1,300	1,040	9.94
Wood Chips	0.5 t	450	225	2.15
Graphite Electrode	0.08 t	12,000	960	9.18
Electricity	12,000 kWh	0.35	**4,200**	**40.15**
Labor			400	3.82
Other Expenses			800	7.65
Environmental Fees			200	1.91
Total Cost (Incl. Tax)			**10,460**	

Dry Season

Table 2.4: Cost Structure of Industrial Silicon in Southwest Dry Season

Cost Component	Unit Consumption	Unit Price (Yuan)	Cost (Yuan)	Share (%)
Direct Raw Materials			**4,860**	**38.51**
Silica	2.5 t	550	1,375	10.90
Coke	0.7 t	1,800	1,260	9.98
Petroleum Coke	0.8 t	1,300	1,040	8.24
Wood Chips	0.5 t	450	225	1.78
Graphite Electrode	0.08 t	12,000	960	7.61
Electricity	12,000 kWh	**0.53**	**6,360**	**50.40**
Labor			400	3.17
Other Expenses			800	6.34
Environmental Fees			200	1.58
Total Cost (Incl. Tax)			**12,620**	

2.2.3 Conclusion

The cost assessment in this report is based on a combination of engineering estimates and market intelligence. Specifically, unit consumption figures (e.g., electricity, silica, reductants) are drawn from publicly available feasibility studies, environmental impact assessment (EIA) documents, and official project disclosures, which provide reliable benchmarks for material intensity under standard operating conditions. Meanwhile, unit prices for key inputs are inferred from a triangulation of information across major commodity data providers such as SMM (Shanghai Metals Market), industry sell-side research, and verified WeChat publications. While these figures are subject to estimation error, they reflect a reasonable approximation of production economics across regions and firms.

According to this analysis, average full-cost estimates suggest that production costs in Xinjiang typically range between 9,000∼10,000 Yuan/ton, while producers in Southwest China (e.g., Yunnan, Sichuan) face higher costs, estimated at 9,500∼13,000 Yuan/ton depending on seasonality and plant efficiency. In contrast, leading firms such as Hoshine Silicon are able to maintain cost levels under 8,500 Yuan/ton through economies of scale, vertical integration, and superior energy arrangements.

In addition to the Northwest and Southwest clusters, several secondary production regions—including Fujian, Guizhou, and Hunan—continue to contribute meaningfully to national industrial silicon output, despite lacking the cost advantages of leading regions. These areas maintain resilience through distinct competitive advantages: Fujian, with its proximity to coastal ports, is oriented toward meeting export demand; Guizhou and Hunan, on the other hand, focus primarily on serving stable, localized downstream demand within their respective regions. Furthermore, although industrial silicon is not a technology-intensive product, differentiation based on impurity control, especially for higher-grade categories such as 3301# and 2202#, remains a key source of value. Certain non-core facilities in these regions specialize in producing grades with tighter specifications on trace elements such as Fe, Al, and Ca, supporting their continued operation and strategic importance as a flexible buffer to mitigate seasonal or policy-induced disruptions in national supply.

2.3 Electricity Cost Control

2.3.1 Captive Power Plants: Lowering Tariff through Vertical Integration

Electricity cost constitutes the single largest component of industrial silicon production, accounting for approximately 35∼45% of total cost. In response, many producers, particularly in Northwest China, have adopted the use of captive power plants to gain greater control over electricity expenses. These facilities are typically coal-fired and built adjacent to submerged arc furnace installations, forming an integrated production chain that reduces reliance on grid electricity.

The key advantage of captive power lies in its significantly lower tariff structure. While the average on-grid industrial electricity price ranges between 0.30∼0.40 Yuan/kWh depending on region and season, captive plants can supply electricity at **0.18∼0.26 Yuan/kWh**, translating into **cost savings of 700∼1,400 Yuan per ton** of industrial silicon produced, assuming a standard consumption of 11,500∼12,000 kWh/t. These savings can materially improve profit margins and serve as a critical differentiator, especially under weak pricing conditions.

However, the implementation of captive power also introduces regulatory and capital complexity. Coal-fired self-generation is subject to increasingly stringent environmental oversight, including local emission quotas, and may not be available in some provinces. Nonetheless, for provinces like Xinjiang and Inner Mongolia, where resource endowment, land availability, and policy incentives align, captive power remains a dominant cost-reduction strategy.

2.3.2 Waste Heat Recovery: Enhancing Energy Efficiency at the Source

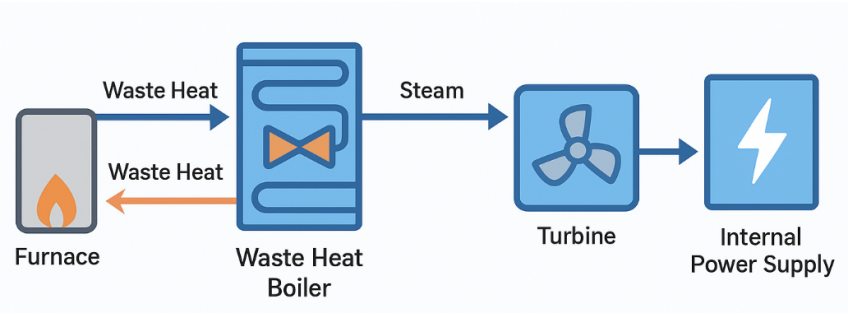


Figure 2.1: Flow of Waste Heat Recovery Process

Waste heat recovery (WHR) has become an increasingly important cost control strategy in industrial silicon production. The process captures residual thermal energy from furnace off-gases and converts it into electricity through steam generation and turbine power systems. While WHR does not cover total electricity demand, it can substantially reduce grid electricity purchases and carbon emissions, especially for large-scale producers.

According to multiple EIA reports from silicon producers in regions like Inner Mongolia, Sichuan, and Yunnan, the deployment of waste heat power generation systems has become increasingly standardized. For instance, one plant with 2×20MW capacity and a rated boiler output of 22 tons/hour achieved a recovery rate of 24%, supplying 297.79 million kWh out of a total 1.24 billion kWh electricity consumption. Another project in Southwest China reported annual electricity savings of 113 million kWh through waste heat reuse, representing over 20% of its on-site power needs. In another case, waste heat power supply contributed 19.98% of the site's total calculated energy input. These figures demonstrate that a well-designed system can consistently offset roughly **20%** of total energy consumption, offering substantial economic and environmental returns.

Despite the clear energy and cost benefits, waste heat recovery remains underutilized across the industry. The primary challenges include high upfront capital investment, ongoing maintenance costs, and the need for stable load factors to ensure recovery system efficiency. Nonetheless, as environmental regulations tighten and producers seek more sustainable cost reductions, waste heat recovery is expected to become a key focus of capacity upgrade and green transition efforts.

2.3.3 Electricity Price Sensitivity Assessment

To better understand the exposure of industrial silicon production cost to electricity-related variables, I conduct a two-dimensional sensitivity analysis based on electricity price (Yuan/kWh) and unit electricity consumption (kWh/t). This allows us to evaluate how changes in either factor would affect total production cost under realistic scenarios.

I used the cost structure of Hoshine's Shanshan, Xinjiang facility as the baseline: 11,000 kWh/t at 0.28 Yuan/kWh results in a total cost of 8,300 Yuan/t. By perturbing the

electricity price in the range of 0.25–0.35 Yuan/kWh and electricity usage in the range of 10,000∼13,000 kWh/t, I constructed a sensitivity matrix. The values represent the total production cost under each scenario. The corresponding heatmap is shown in Figure 2.2.

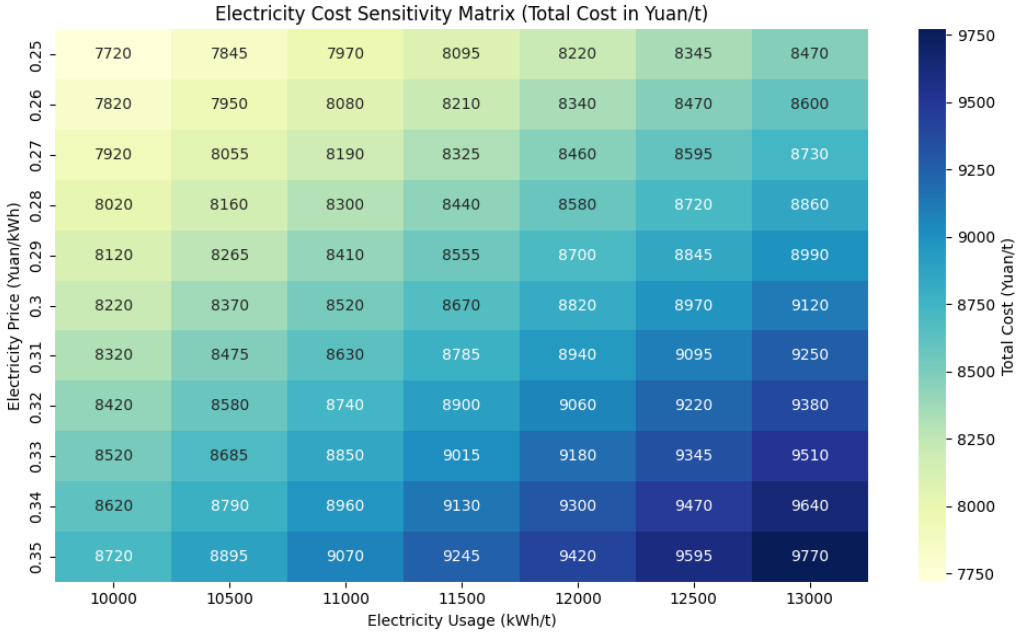


Figure 2.2: Electricity Cost Sensitivity Matrix (Yuan/t)

I also calculated elasticity to assess the marginal sensitivity. When electricity price increases from 0.28 to 0.30 Yuan/kWh at constant 11,000 kWh/t usage, the estimated elasticity of total cost with respect to electricity price is:

$$\varepsilon = \frac{\%\Delta \text{Cost}}{\%\Delta \text{Price}} = \frac{(8520 - 8300)/8300}{(0.30 - 0.28)/0.28} \approx 0.371$$

This suggests that each 1% increase in electricity price leads to approximately a 0.37% increase in total cost under current conditions.

2.4 Rough Sketch of Cost-Capacity Curve across Representative Producers

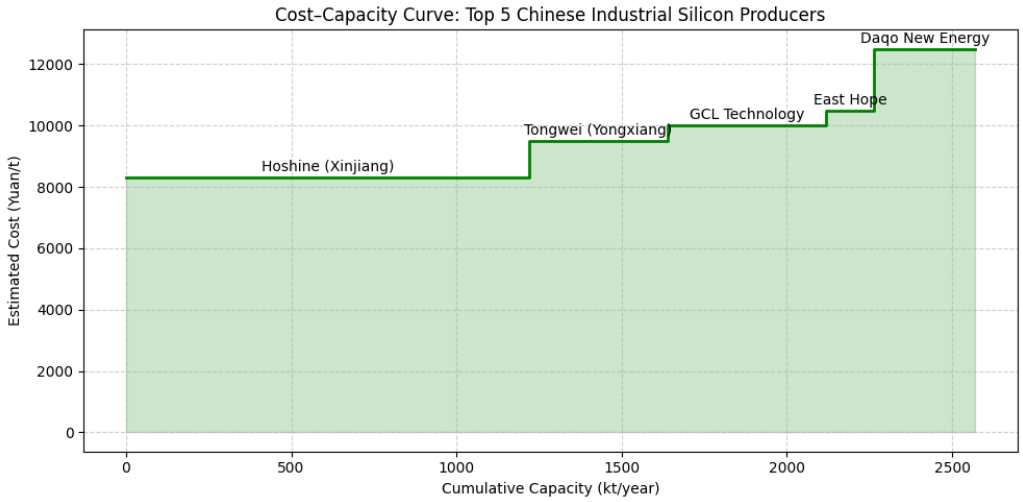


Figure 2.3: Cost-Capacity Curve for 5 Largest Producers in China

Chapter 3

Downstream Demand Structure

3.1 Overview of Downstream Demand

Industrial silicon serves as a foundational input across several major downstream sectors, including **polysilicon** for photovoltaics, **organosilicon compounds** for chemical materials, and **aluminum-silicon alloys** widely used in the automobile and real estate industries. Over the past five years, the downstream composition of demand has undergone a significant transformation: the share of industrial silicon consumed by the polysilicon industry surged from approximately **19% in 2019** to **49% by 2024**, driven by explosive growth in photovoltaic capacity additions. Meanwhile, the demand share of organosilicon has remained relatively stable at around **30∼40%**, reflecting its mature role in coatings, sealants, and industrial chemicals. In contrast, aluminum alloy applications have declined in relative importance, with their share dropping from **over 40% to about 16%** during the same period, due to slower growth in traditional sectors like automotive casting and real estate construction.

These shifts have not only altered the volume distribution of industrial silicon demand, but also reshaped the required product structure. Different applications call for distinct silicon grades (553# for foundry and deoxidizer use, 421#/3303#/2202# for chemical synthesis, and ultra-pure inputs for solar-grade polysilicon), created differentiated cost structures and supply chain pathways. Looking ahead, the demand trajectory will likely continue to favor high-purity, integrated applications, reinforcing the strategic importance of grade-specific supply matching and upstream–downstream coordination.

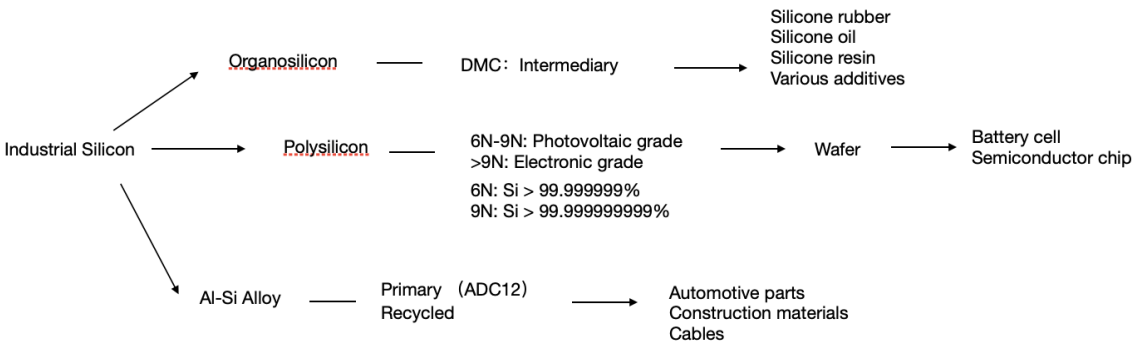


Figure 3.1: Demand Overview

3.2 Major Demand Sectors

3.2.1 Polysilicon: A Photovoltaic-Driven Growth Engine

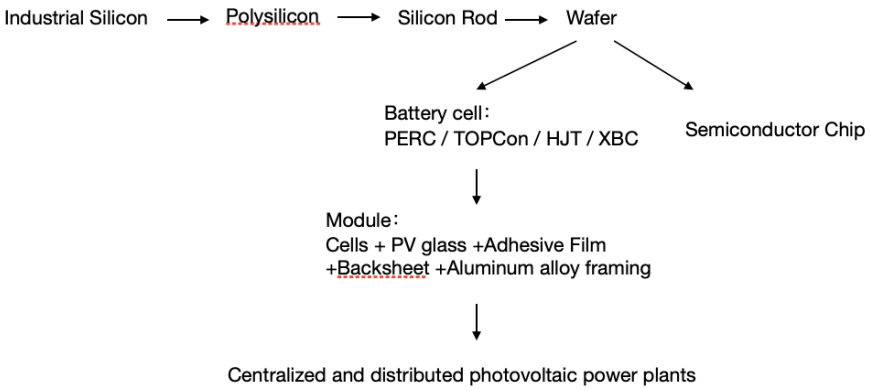


Figure 3.2: Polysilicon Industry Chain

Polysilicon, or polycrystalline silicon, is the foundational raw material for photovoltaic (PV) and semiconductor industries. Due to its semiconducting properties, high abundance, and mature purification technology, silicon has become the dominant material for solar power generation. Its stable crystal structure enables the conversion of light into electricity through the photovoltaic effect, making it the primary feedstock in over 95% of global solar cells.

Polysilicon used in photovoltaics is typically classified by purity: solar-grade material (6N∼9N) and electronic-grade material (9N and above). Further downstream, polysilicon is drawn into monocrystalline silicon rods, sliced into wafers, and eventually processed into solar cells or semiconductor chips. While the theoretical silicon-to-polysilicon transformation is nearly 1:1 by weight, actual unit consumption is affected by deposition efficiency, recycling rate, and quality standards of the end product. Industry-standard unit consumption is generally between 1.05∼1.15 kilograms of industrial silicon per kilogram of polysilicon.

Two primary production routes dominate the market: the modified Siemens process and the fluidized bed reactor (FBR) process for granular silicon. The former offers high purity and stability, making it the mainstream technology, whereas the latter is gaining ground for its lower energy consumption and faster production cycle. As the end of 2024, global operational polysilicon capacity reached 3.249 million tons, with effective capacity at 3.394 million tons, an increase of 38.1% year-over-year. Actual global output reached approximately 1.957 million tons, representing a 21.7% year-on-year increase. Among these, solar-grade chunk polysilicon contributed 1.632 million tons (83.4%), granular silicon 286,000 tons (14.6%), and electronic-grade polysilicon 39,000 tons (2.0%).

Production in this segment is highly concentrated. In 2024, the top 10 global producers accounted for 82.2% of capacity and 86.9% of output, with all of the top 5 located in China. These five firms alone produced 1.3758 million tons, representing 70.3% of global output.

A key structural shift is the growing demand for high-purity N-type polysilicon, used for N-type wafers that enable higher photovoltaic conversion efficiency. While P-type polysilicon typically meets solar-grade or electronic-grade level 3 standards, N-type material demands electronic-grade level 2 or above, requiring denser, more uniform morphology. Although both share similar process routes, the performance requirements for N-type are more stringent. According to data from the China Photovoltaics Industry Association (CPIA), the market share of N-type polysilicon surpassed 60% by May 2024, and leading producers report N-type output ratios exceeding 80%.

Demand Estimation

There are two mainstream approaches to estimate industrial silicon demand from the polysilicon segment: a top-down method based on photovoltaic installation volumes, and a production-driven method based on actual polysilicon output.

- **Top-down method**: Starting from photovoltaic module demand (typically reported in new installations in GW), one can backtrack through the value chain by applying **theoretical material transmission ratios**. These ratios capture silicon consumption per unit of electricity generation capacity (in g/W), and depend on cell type and conversion efficiency. Based on prevailing technologies, the range of silicon consumption per watt typically falls between **1.60 and 2.02 g/W**. Thus, demand for industrial silicon can be indirectly inferred from new PV installation data.

- **Production-driven method**: Alternatively, monthly polysilicon output serves as a more direct proxy. Given the mass-based transformation between industrial silicon and polysilicon, taking processing loss and processing inefficiencies into account, the effective unit consumption typically ranges from **1.04 to 1.20 kilograms of industrial silicon per kilogram of polysilicon**. Since this upstream conversion chain (industrial silicon $\rightarrow$ polysilicon) is technologically unified and industrially centralized, actual polysilicon production is a robust indicator of corresponding silicon demand.

As shown in Table 3.1, China's polysilicon output in the first half of 2025 stabilized around **90∼100 thousand tons** per month. Despite year-on-year declines exceeding 40%, this was mainly a reflection of the high base effect in 2024, rather than structural deterioration in demand.

Table 3.1: Monthly Polysilicon Output and Year-on-Year Change (2025 H1)

Month	**Output (10,000 tons)**	**YoY (%)**	Est. Si Demand (1.08 t/t)
Jan	9.44	-40.59	10.20
Feb	9.01	-44.72	9.73
Mar	9.61	-43.80	10.38
Apr	9.54	-47.35	10.30
May	9.61	-45.71	10.38
Jun	10.10	-33.11	10.91

3.2.2 Organosilicon

Organosilicon compounds represent a critical downstream application of industrial silicon, forming a vast family of silicon-carbon bonded chemicals widely used across the construction, electronics, medical, automotive, and personal care industries. These materials are prized for their exceptional thermal stability, water repellency, chemical inertness, and electrical insulation properties. The production of organosilicon begins with the synthesis of organosilicon monomers, among which **dimethyldichlorosilane ($\mathbf{Si(CH_3)_2Cl_2}$)** is the primary intermediate. This compound is manufactured through the Müller-Rochow process, which reacts chloromethane with silicon in the presence of a copper catalyst:

$$2CH_3Cl + Si \rightarrow Si(CH_3)_2Cl_2$$

Subsequent hydrolysis and polymerization of these monomers yield cyclic compounds such as **dimethylcyclosiloxane (DMC)**, expressed chemically as $[(CH_3)_2SiO]_n$ where n=3∼6, which serve as the feedstock for most silicone rubbers, fluids, and resins:

$$Si(CH_3)_2Cl_2 \rightarrow [(CH_3)_2SiO]_n$$

By 2025, the domestic production of DMC reached a monthly peak of approximately 22.07 kilotons, with a half-year cumulative total of 121.42 kilotons, indicating sustained activity in the organosilicon segment. The overall industry operates with a relatively dispersed end-use landscape, lacking the centralized scale of photovoltaics, yet maintains a stable baseline of consumption driven by construction sealants, personal care products, and electronics insulation materials. Margins and operating rates are also influenced by electricity prices and environmental policies in regions such as Yunnan and Shandong.

Demand Estimation

Unlike the photovoltaic industry, where final demand can be projected from power station installations, the fragmented nature of organosilicon consumption precludes end-use forecasting. Instead, industry consensus relies on tracking **DMC output** as a proxy for silicon demand. The chemical transformation involved offers a calculable conversion factor: producing 1 ton of DMC requires approximately **0.52 tons** of industrial silicon, derived as follows:

- 1 ton DMC ⇐ 2 tons organosilicon monomer (primarily $Si(CH_3)_2Cl_2$)
- 1 ton monomer ⇐ 0.26 tons of industrial silicon
- Therefore, 1 ton DMC ⇐ 0.52 tons of industrial silicon

In practice, unit consumption varies depending on process efficiency and catalyst systems:

- Advanced processes: 0.47∼0.49 tons Si/ton DMC
- Industry average: 0.52∼0.60 tons Si/ton DMC
- Traditional processes: up to 0.60 tons Si/ton DMC

Table 3.2: Monthly DMC Output and Year-on-Year Change (2025 H1)

Month	**Output (10,000 tons)**	**YoY (%)**	Est. Si Demand (0.52 t/t)
January	22.07	15.13	11.48
February	19.95	11.76	10.37
March	18.79	-3.84	9.77
April	17.28	-7.25	8.99
May	18.40	-11.33	9.57
June	20.93	1.60	10.88

3.2.3 Aluminum-Silicon Alloys

Aluminum-silicon (Al-Si) alloys represent a critical downstream segment of industrial silicon, forming the structural backbone of many modern industrial systems. They are widely applied in automotive, construction, photovoltaic, and transportation industries, thanks to their favorable combination of light weight, corrosion resistance, thermal conductivity, and castability.

There are two major dimensions for categorizing Al-Si alloys:

- **Processing route**:
  - **Cast aluminum alloys** (e.g., ADC12, A356, A380) are produced via molten casting and have high silicon content (typically 7%–13%) to improve fluidity and reduce shrinkage defects. These are dominant in automotive and machinery parts.
  - **Deformed aluminum alloys** (e.g., 4***, 6*** series) are shaped via rolling or extrusion. While their silicon content is generally lower (0.2%∼1.7%).
- **Material origin**:
  - **Primary aluminum alloys** are produced from electrolytic aluminum and require fresh alloying materials, including industrial silicon.
  - **Recycled (secondary) aluminum alloys** are derived from scrap and usually retain pre-existing alloying elements, often requiring less or even no additional silicon.

Al-Si alloys are foundational in:

- **Automotive manufacturing**, especially for internal combustion engines (ICE, including engine blocks, cylinder heads, pistons) and new energy vehicles (motor housings, battery trays). ADC12 and A356 are key casting grades, while 6061 and 6063 dominate deformed applications.
- **Construction and real estate**, where 6xxx series alloys are used in door/window frames, curtain walls, and aluminum formwork systems.
- **Photovoltaics**, where extruded Al-Si profiles (mostly 6063) serve as module frames and support brackets.

With the ICE market in gradual decline, demand for Al-Si-Cu alloys used in traditional engines is falling. Conversely, the electrification of vehicles and rapid expansion of solar installations are propelling the consumption of high-silicon aluminum alloys, especially those in the Al-Si and Al-Mg-Si systems.

Demand Estimation

Given that recycled aluminum alloys generally do not require significant silicon additions, demand estimation focuses on **primary aluminum alloy production**. Based on industry standards and metallurgical balances, an average of: **0.22 tons of industrial silicon are consumed per ton of primary Al-Si alloy**.

This unit consumption reflects weighted silicon content across the major grades in use, accounting for both casting and deformation alloys.

Table 3.3: Monthly Primary Aluminum Alloy Output and Estimated Silicon Demand (2025 H1)

Month	**Alloy Output (10,000 tons)**	**YoY (%)**	Est. Si Demand (0.22 t/t)
Jan	25.0	0.20	5.50
Feb	24.0	0.28	5.28
Mar	26.8	0.13	5.90
Apr	26.2	0.02	5.76
May	26.1	0.04	5.74
Jun	25.5	-0.02	5.61

Primary aluminum alloy output has remained **highly stable in recent months**, with year-on-year growth rates fluctuating narrowly between **-0.02% and +0.28%**. This stability underscores a relatively mature demand structure.

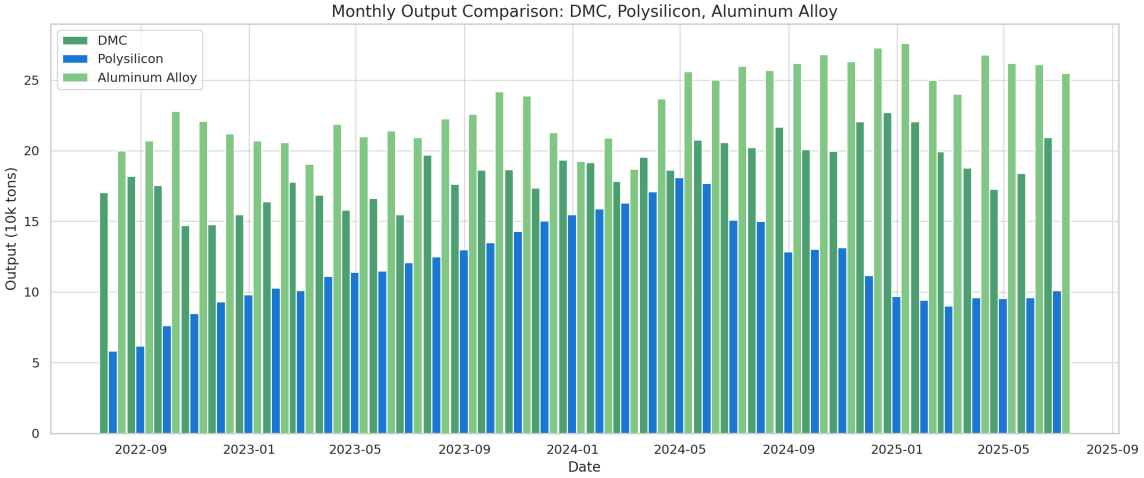


Figure 3.3: Monthly Output Comparison - Three Segments

Chapter 4

Forecasting Industrial Silicon Futures Prices: Model Selection, Evaluation, and Interpretation

4.1 Overview

This chapter presents a forecasting study of industrial silicon futures prices using historical market data. The primary objective is to evaluate the feasibility and predictive performance of classical time-series models in capturing short-term price dynamics within this commodity market. Specifically, the analysis aims to forecast the closing price of industrial silicon futures for the subsequent seven trading days.

The dataset employed in this study consists of authorized historical futures statistics, including daily closing prices, trading volumes, and open positions. These variables collectively reflect both market activity and investor sentiment, providing a comprehensive foundation for model construction.

Data and Variable Selection The analysis focuses on the Active contract, defined as a smoothed series that tracks the price trajectory of the most active futures contracts over time. This smoothing procedure minimizes discontinuities caused by contract rollovers, thereby generating a continuous and representative time series suitable for modeling.

In addition to the closing price, trading volume and open interest of the contracts are incorporated as potential exogenous regressors in certain models. Trading volume captures the intensity of market participation, while open interest reflects traders' overall confidence and exposure in the market. Including these variables allows the forecasting framework to account for both price-level dynamics and behavioral aspects of market activity.

Modeling Objective By applying and comparing several time-series forecasting models, this study seeks to determine whether industrial silicon futures prices exhibit sufficient structure and temporal dependence to be reliably predicted using standard modeling techniques.

4.2 Model Selection

4.2.1 Candidate Models

Autoregressive Integrated Moving Average (ARIMA)

ARIMA models the current value of a time series as a linear combination of its **own past values** (autoregressive part), **past forecast errors** (moving average part), and differencing operations to ensure stationarity. The general form is ARIMA(p, d, q), where p is the autoregressive order, d is the degree of differencing, and q is the moving average order.

- Assumptions
  1. Stationarity of the series after appropriate differencing.
  2. Linearity between past and present values.
  3. Residuals are uncorrelated, homoskedastic, and normally distributed.
- Advantages and Limits
  1. Simple and interpretable.
  2. Performs well on short-term, univariate forecasting.
  3. Sensitive to non-stationarity and parameter specification.
  4. Not ideal when exogenous predictors or regime changes are important.
  5. Cannot easily capture nonlinearities.

Seasonal ARIMA (SARIMA)

SARIMA extends ARIMA to include seasonal components, suitable when the series exhibits periodic patterns (e.g., monthly or quarterly cycles). The model is denoted as SARIMA(p, d, q)(P, D, Q)$_s$, where s is the seasonal period.

- Assumptions
  1. Presence of a stable seasonal pattern.
  2. The series is stationary after appropriate differencing.
- Advantages and Limits
  1. Captures both short-term and seasonal dynamics.
  2. Maintains ARIMA interpretability.
  3. Parameter estimation can become complex and unruly with many hyperparameters, likely to overfit.
  4. Less effective for data with irregular or shifting seasonality.

Exponential Smoothing (ETS)

ETS methods forecast by applying exponentially decaying weights to past observations. They model trend and seasonality explicitly via smoothing equations rather than autoregressive structures. Although this model is the most intuitive and handy one, but it is not suitable for long-term or highly volatile data like futures prices.

4.2.2 Preliminary Analysis and Model Selection

Visual Inspection Figure 4.1 displays the daily closing price of the active industrial silicon futures contract. The price series exhibits a clear downward trend from early 2023 to mid-2025, interspersed with several periods of partial recovery. This nonstationary behavior suggests that differencing will be required before applying most classical time-series models such as ARIMA or VAR.

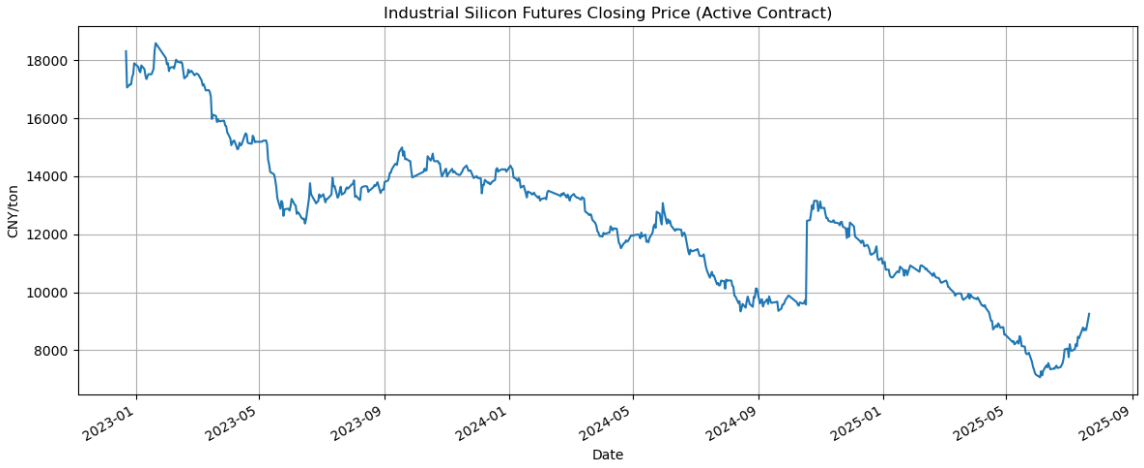


Figure 4.1: Futures Closing Price (Active Contract)

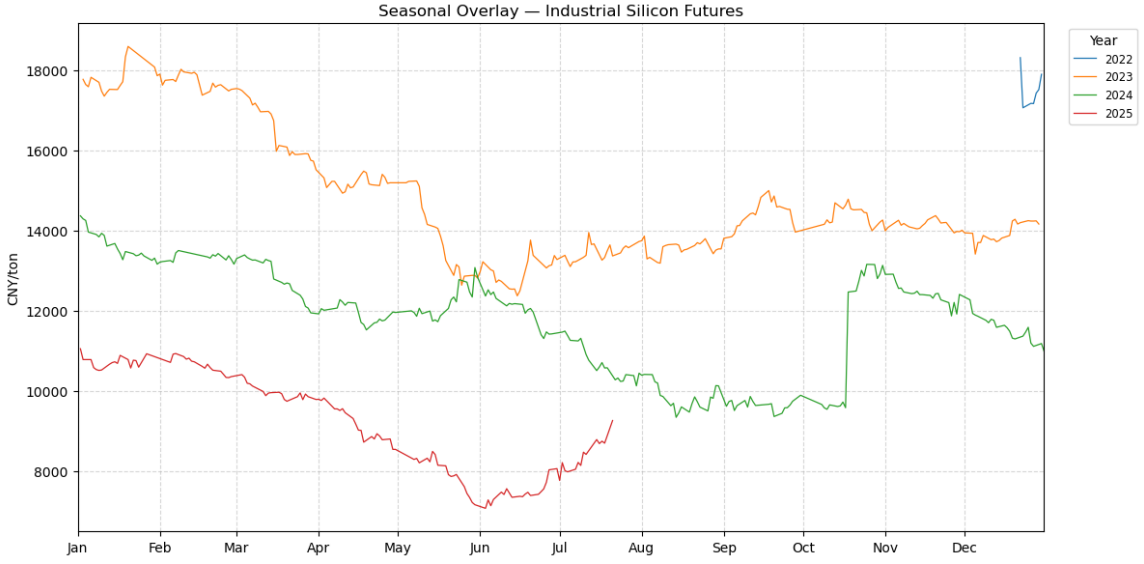


Figure 4.2: Futures Closing Price (Active Contract) Seasonality

Stationarity Assessment To formally assess stationarity, the series was differenced once, and an Augmented Dickey–Fuller (ADF) test was conducted (Figure 4.3). The resulting p-value indicates that the first-differenced series is stationary, satisfying one of the core assumptions for linear time-series models. The differenced plot also shows relatively stable variance, aside from isolated spikes that may correspond to short-term market shocks or contract rollover effects.

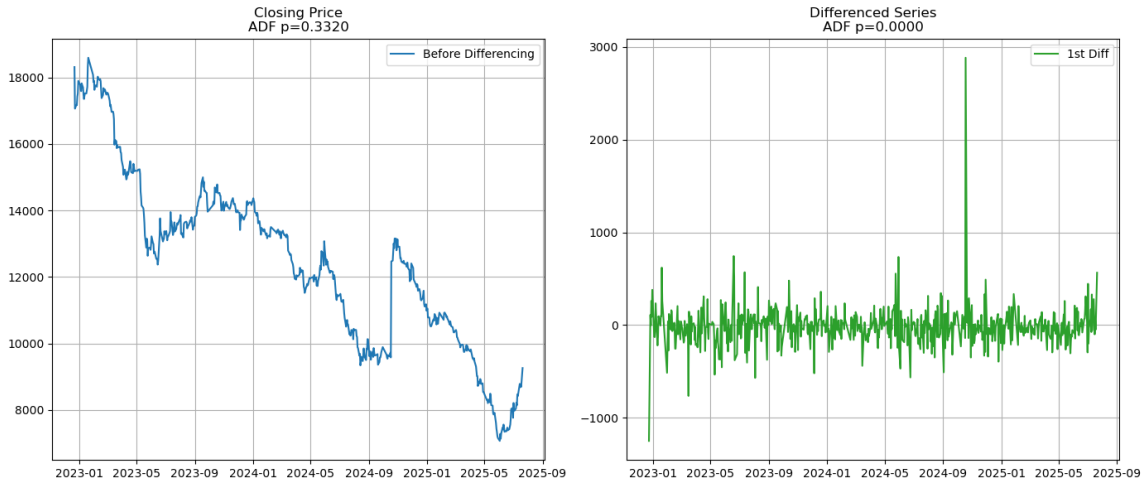


Figure 4.3: Stationary of Raw Series and Lag[1]

Volatility Assessment To evaluate the presence of conditional heteroskedasticity in the industrial silicon futures returns, an ARCH–LM test was performed on the residual series obtained after first differencing. The test statistics (LM = 0.0663, F = 0.0065) with corresponding p-values of 1.0000 indicate a failure to reject the null hypothesis of homoskedasticity. This suggests that, over the sample period, volatility does not exhibit statistically significant clustering at conventional confidence levels. In other words, the conditional variance appears relatively stable, and the short-term return dynamics are not dominated by persistent volatility shocks. Therefore, the GARCH model is not appropriate in this scenario.

Preliminary Implications for Model Selection

- Price dynamics are dominated by medium-term market cycles and external shocks (e.g. electricity price variation or capacity policy) rather than by calendar-driven seasonality (Figure 4.2).
- Any model for conditional mean dynamics should incorporate at least one order of differencing ($d = 1$), making ARIMA family models natural candidates.

4.3 ARIMA Model

4.3.1 ARIMA Assumption Checking

Before estimating the ARIMA model, the differenced series was examined to ensure that it satisfied the fundamental assumptions of stationarity and linear dependence. The Augmented Dickey–Fuller (ADF) test (Figure 4.3) confirmed that the first-differenced price series was stationary, allowing the use of an integrated model with d=1.

Autocorrelation and partial autocorrelation plots were then analyzed to determine the appropriate lag structure (Figure 4.4). The autocorrelation function (ACF) displayed a single significant spike at lag 1 before rapidly decaying to within the confidence bounds, while the partial autocorrelation function (PACF) showed a similar cutoff pattern. This

behavior is characteristic of a first-order moving average process. Consequently, an ARIMA(0,1,1) model was selected as a parsimonious representation of the differenced series, consistent with the pattern of short-term memory observed in the data.

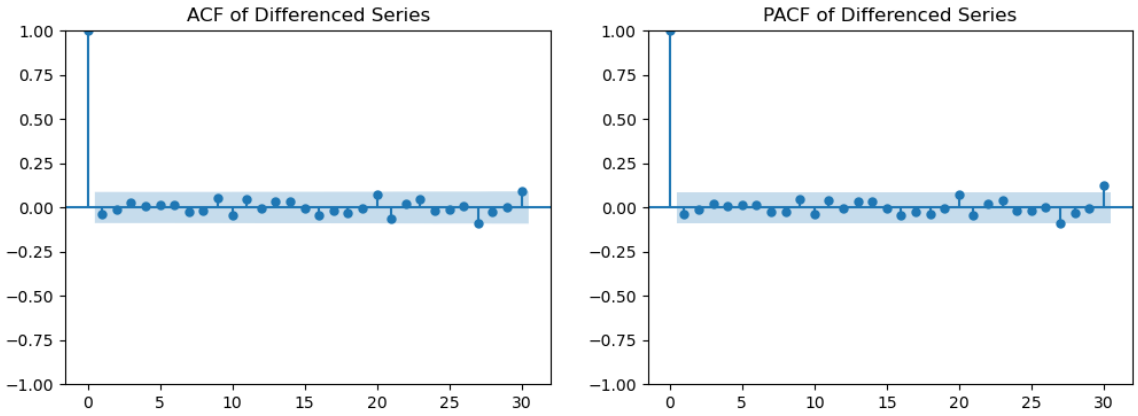


Figure 4.4: ACF and PACF Test

The automated model selection procedure using the `auto_arima` function from `pmdarima` corroborated this conclusion, identifying ARIMA(0,1,1) with an intercept as the best-fitting model according to the Akaike Information Criterion (AIC). (Figure 4.5)

```
Performing stepwise search to minimize aic
 ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=6809.503, Time=0.02 sec
 ARIMA(1,1,0)(0,0,0)[0] intercept   : AIC=6808.010, Time=0.04 sec
 ARIMA(0,1,1)(0,0,0)[0] intercept   : AIC=6807.969, Time=0.03 sec
 ARIMA(0,1,0)(0,0,0)[0]             : AIC=6809.855, Time=0.00 sec
 ARIMA(1,1,1)(0,0,0)[0] intercept   : AIC=6809.964, Time=0.05 sec
 ARIMA(0,1,2)(0,0,0)[0] intercept   : AIC=6809.963, Time=0.04 sec
 ARIMA(1,1,2)(0,0,0)[0] intercept   : AIC=6811.962, Time=0.06 sec
 ARIMA(0,1,1)(0,0,0)[0]             : AIC=6808.645, Time=0.01 sec

Best model:  ARIMA(0,1,1)(0,0,0)[0] intercept
Total fit time: 0.244 seconds
```

Figure 4.5: Result of AutoARIMA

4.3.2 Training and Testing Subset Split

To evaluate the out-of-sample predictive performance of the ARIMA model, the time series was divided into a training and testing subset following an 80:20 ratio. The training period spans from **22 December, 2022 to 10 January, 2025**, covering the majority of market fluctuations and providing sufficient information for parameter estimation. The testing period extends from **13 January, 2025 to 21 July, 2025**, representing the most recent six months of observations, during which model forecasts are compared against realized prices.

This temporal split preserves the chronological order of observations to avoid look-ahead bias and to mimic a realistic forecasting environment. The training data are used exclusively for model estimation and diagnostic testing, while the testing data serve to

assess forecast accuracy based on root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). This design ensures that model evaluation reflects true predictive capability rather than in-sample fit.

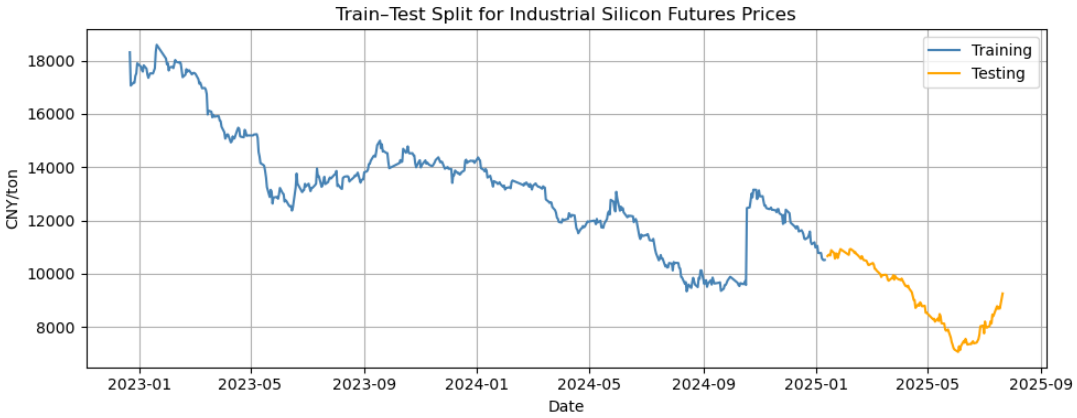


Figure 4.6: Split Data Into Training and Testing Subsets

4.3.3 Fitted Model

Following the assumption verification, the ARIMA(0,1,1) model was fitted to the training sample using maximum likelihood estimation.

SARIMAX Results

Dep. Variable:	Futures Closing Price (Active Contract): Industrial Silicon	**No. Observations:**	498
Model:	ARIMA(0, 1, 1)	**Log Likelihood**	-3402.322
Date:	Thu, 09 Oct 2025	**AIC**	6808.645
Time:	10:40:14	**BIC**	6817.062
Sample:	0	**HQIC**	6811.949
	- 498		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
ma.L1	-0.0808	0.033	-2.446	0.014	-0.145	-0.016
sigma2	5.204e+04	671.751	77.473	0.000	5.07e+04	5.34e+04

Ljung-Box (L1) (Q):	0.81	**Jarque-Bera (JB):**	60946.68
Prob(Q):	0.37	**Prob(JB):**	0.00
Heteroskedasticity (H):	1.62	**Skew:**	3.94
Prob(H) (two-sided):	0.00	**Kurtosis:**	56.67

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Figure 4.7: ARIMA Fitted Model

The model includes a single first-order moving average term and one order of differencing, as identified during the assumption checking stage.

Model Diagnostics

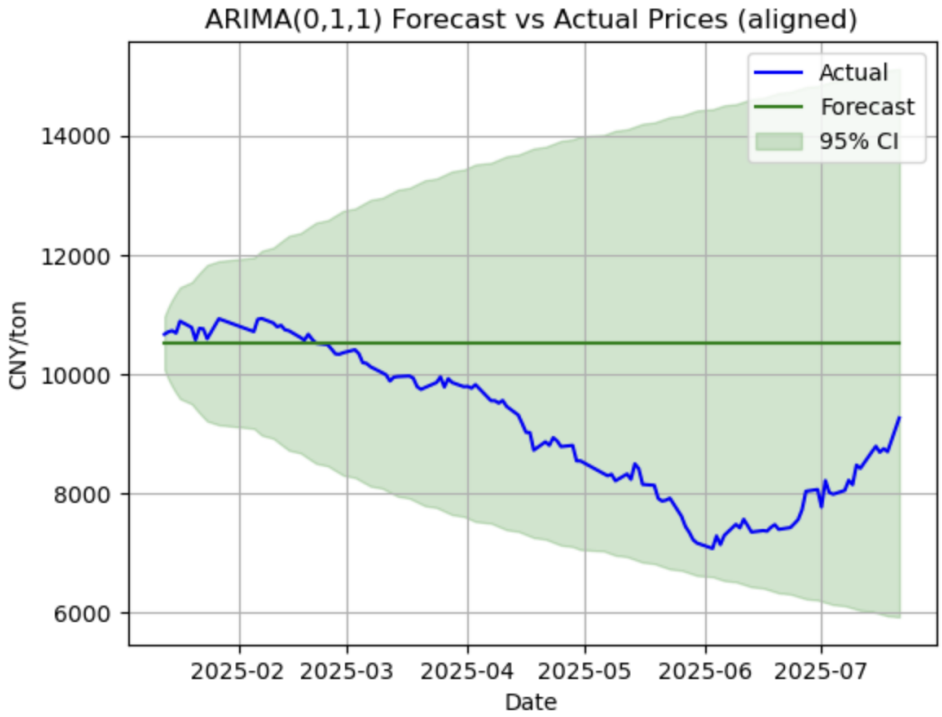
The chosen model passed standard diagnostic checks, confirming its adequacy for modeling short-term dynamics in industrial silicon futures prices. The Ljung–Box Q-statistic at lag [1] (Q = 0.81, p = 0.37) fails to reject the null hypothesis of no residual autocorrelation, indicating that the model effectively captured all linear dependencies within the differenced series. **The residuals therefore approximate white noise**, suggesting that no additional autoregressive or moving-average terms are necessary. Although the Jarque–Bera statistic (JB = 60,946.68, p = 0.00) rejects normality, reflecting the **heavy tails and skewness typical of commodity price data, this deviation does not compromise the model's validity for forecasting purposes**. The heteroskedasticity test (H = 1.62, p = 0.00) indicates mild variance variation across the sample, but not to a degree warranting volatility modeling extensions such as GARCH, consistent with earlier ARCH–LM results.

Results and Interpretation

The estimated moving-average coefficient (MA(1) = –0.0808, p = 0.014) is statistically significant, suggesting a small but meaningful short-run adjustment mechanism in daily price changes. This negative coefficient implies that a positive shock to prices is partially offset by a minor correction in the subsequent period. The estimated innovation variance ($\sigma^2 = 52,040$) reflects the typical magnitude of daily price fluctuations in the active contract. **Together, these findings indicate that industrial silicon futures prices follow a near-random-walk process, with minimal predictable structure beyond first-order noise smoothing.** This behavior aligns with theoretical expectations for efficient commodity markets, where new information is rapidly incorporated into prices, leaving only minor short-term autocorrelation.

Forecast Evaluation on Testing Sample

The predictive performance of the chosen model was evaluated using the hold-out testing sample. Figure 4.8 illustrates the model's static, multi-step forecasts, while Figure 4.9 presents the corresponding rolling one-step-ahead forecasts.



RMSE: 1866.61
MAE: 1497.75
SMAPE: 16.03%

Figure 4.8: Model Performance on Testing Subset

Since ARIMA(0,1,1) can be expressed as an MA(1) process on the differenced series, its long-horizon expectation converges to the last observed value, implying that future changes are treated as white noise with zero mean. In other words, the model assumes that price variations are unpredictable beyond the short run, leading to a constant mean forecast accompanied by widening confidence intervals.

While the forecast mean remains flat, it is notable that the actual test-period observations fall completely within the 95% confidence interval. This indicates that, despite the model's inability to anticipate turning points or short-term fluctuations, it still provides a statistically valid estimate of the overall price level. The bounded forecast range effectively captures the uncertainty around the long-run equilibrium implied by the ARIMA(0,1,1) structure, suggesting that the model's predictive distribution is well-calibrated.

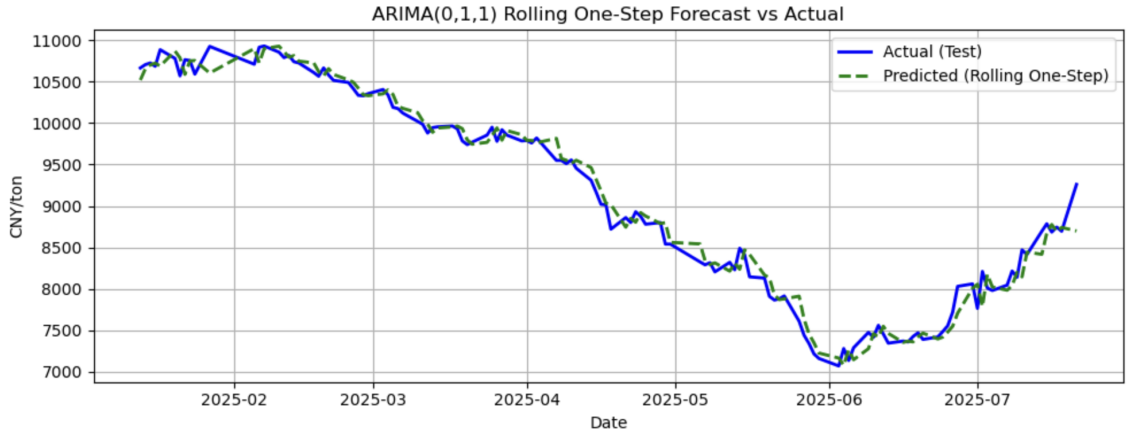


Figure 4.9: Model Performance on Testing Subset Rolling One-Step-Ahead

In contrast, the rolling one-step-ahead forecast demonstrates strong tracking accuracy, as the model updates sequentially using newly observed data at each step. The predicted series closely follows the actual trajectory of industrial silicon futures prices, successfully capturing both the mid-sample decline and the subsequent recovery in mid-2025. This result confirms that the ARIMA(0,1,1) model, while limited in its multi-step forecasting ability, remains effective for short-horizon iterative forecasts when re-estimated in real time. Overall, the testing-sample evaluation supports the model's adequacy for near-term price prediction, though the flat long-horizon forecast underscores the absence of persistent trend or autocorrelation structure in the underlying data.

4.4 Discussion

4.4.1 Achievement and Limitations

Achievement The implementation of the ARIMA(0,1,1) model on industrial silicon futures data provides several valuable insights into the practical application of time-series forecasting. First, this analysis deepened the understanding of how an ARIMA framework behaves when applied to real-world, high-frequency financial data. By following the complete workflow (i.e. from stationarity assessment and model selection to diagnostic evaluation and forecast validation) the study demonstrated how differencing can effectively stabilize nonstationary price series and how moving-average dynamics can capture short-term noise. Moreover, the model exhibited a degree of predictability, particularly through the rolling one-step-ahead forecasts, which tracked market trends with reasonable accuracy. This result highlights that, even for a relatively simple ARIMA specification, the model can yield informative short-run forecasts when updated regularly with new data.

Limitation The ARIMA(0,1,1) model offers limited guidance for actual investment decision-making, as it primarily models price levels rather than directional or volatility-based trading signals. The flat multi-step forecasts further reveal that, in the absence of newly observed data, the model quickly converges to the last known value, making it unsuitable for anticipating trend reversals or price surges. Additionally, because ARIMA models rely solely on past price information, their predictive power diminishes when

structural shifts, policy changes, or supply–demand shocks occur. The inability to incorporate real-time external information or evolving market fundamentals restricts the model's long-horizon forecasting capability.

4.4.2 Future work

While the ARIMA modeling exercise provided valuable insights into the time-series behavior of industrial silicon futures, several directions remain open for further exploration. One promising avenue is the adoption of more flexible machine learning frameworks to enhance predictive performance. Although alternative ARIMA-family models (including ARIMAX specifications with open interest and trading volume, as well as GARCH variants designed to capture volatility clustering) were tested during this study, none produced substantively better forecasts than the baseline ARIMA(0,1,1) model. This outcome suggests that linear, history-dependent structures may be insufficient to fully capture the complexity of the market dynamics. In contrast, modern machine learning models such as Random Forests, Gradient Boosting, or Long Short-Term Memory (LSTM) networks may uncover nonlinear relationships and interaction effects that traditional econometric approaches overlook. Incorporating these models could offer not only improved accuracy but also a more nuanced understanding of how various predictors jointly influence futures prices.

Beyond methodological innovation, a more fundamental challenge lies in addressing the policy-driven and sentiment-sensitive nature of the industrial silicon market. During the research process, and reinforced by firsthand observations from an internship in the sector, it became apparent that price movements are often detached from immediate supply–demand fundamentals. For example, in July 2025, futures prices surged sharply despite stable capacity levels and subdued demand indicators, largely due to speculative responses to government meetings and public statements about "Anti-Involution". Such events highlight that market participants often trade not on realized fundamentals, but on perceived policy tone and anticipated intervention. Capturing these dynamics empirically would require constructing a political or sentiment index, capable of quantifying the intensity of policy-related signals and their timing relative to price fluctuations. While operationalizing such an index presents challenges (particularly in assigning objective weights to qualitative information) future research could combine text mining, natural language processing (NLP), and event-study methods to systematically measure and model the influence of political communication on market expectations.

Taken together, these extensions would not only refine the predictive capacity of the model but also deepen its interpretive scope, bridging the gap between econometric forecasting and behavioral market analysis. A hybrid framework that integrates quantitative price patterns with qualitative policy sentiment would represent a significant step forward in understanding and anticipating movements in the industrial silicon futures market.